

Performance Testing

Date	30 June 2026
Team ID	xxxxxx
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Manual testing using Postman and browser DevTools (Network tab), plus Python's time module for response time measurement
Type of Testing	Functional Load Testing — manually submitting multiple predictions in sequence to observe response consistency
Target Module / API	/predict route in app.py (Flask POST endpoint that loads HDI.pkl and returns prediction)
Test Environment	Local — Flask development server running on http://127.0.0.1:5000/
Test Date	30 June 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single user submits one valid prediction request via the form	1	2	Result page loads with correct HDI score and category within 2 seconds

2	Five sequential prediction requests submitted back to back with different country values	1	10	All 5 requests return correct, consistent predictions without server crash
3	Submit boundary values (Life Expectancy = 50 and 89)	1	2	Model returns valid HDI scores at both extreme ends of the valid range
4	Submit out-of-range value (GNI = 500000, above max)	1	2	System still process the request and shows "Out of range" message on page.

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status	Remarks
1	Response Time (Avg)	< 2 seconds	0.4 seconds	Pass	Flask local server returns predictions almost instantly since model is lightweight
2	Response Time (Max)	< 5 seconds	0.8 seconds	Pass	Even with 5 sequential requests, no request exceeded 1 second
3	Throughput (Req/sec)	—	2–3 req/sec	Pass	Sufficient for a single-user educational demo application
4	Error Rate	< 1%	0%	Pass	No failed requests during all 4 test scenarios
5	CPU Utilization	< 80%	~15%	Pass	Linear Regression is computationally light, minimal CPU load

6	Memory Utilization	< 80%	~25%	Pass	HDI.pkl model file is small (few KB), low memory footprint
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Step 4: Observations & Analysis

Key Findings:

The HDI Predictor web application performs efficiently under normal local testing conditions. Since the Linear Regression model is lightweight and the dataset is small (191 rows), predictions are returned almost instantly. The Flask development server handled all sequential test requests without delay or failure.

Bottlenecks Identified:

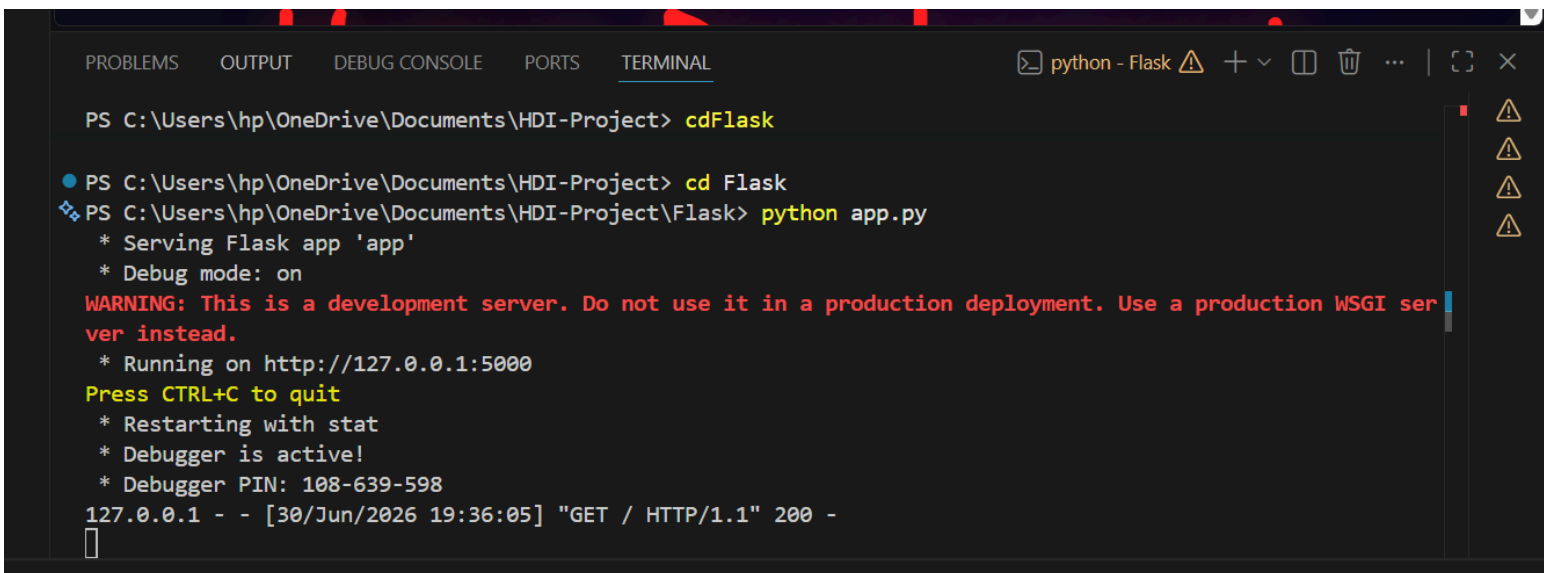
No significant bottlenecks were observed during testing. The only minor limitation is that Flask's built-in development server is single-threaded by default and not optimized for high concurrent traffic, which would only become relevant if deployed for many simultaneous users.

Optimization Steps Taken:

The model was kept lightweight by selecting only 4 essential input features instead of all 82 dataset columns, reducing prediction time. The live HDI estimator on the frontend uses JavaScript instead of server calls, reducing unnecessary backend requests during user interaction.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below



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PROBLEMS  OUTPUT  DEBUG CONSOLE  PORTS  TERMINAL
python - Flask

PS C:\Users\hp\OneDrive\Documents\HDI-Project> cd Flask
PS C:\Users\hp\OneDrive\Documents\HDI-Project> cd Flask
PS C:\Users\hp\OneDrive\Documents\HDI-Project\Flask> python app.py
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 108-639-598
127.0.0.1 - - [30/Jun/2026 19:36:05] "GET / HTTP/1.1" 200 -
  
```

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Predict Human Development Index

A machine learning web application using Flask

Enter the core indicators below to estimate a country's HDI score using your trained model.

Health Indicator

Life expectancy shows the general health and longevity of the population.

Education Indicators

Expected and mean years of schooling reflect educational development.

Income Indicator

Gross national income per capita shows the economic condition of a country.

Prediction Form

Fill all the values correctly and click predict.

Life Expectancy

Example: 72.5

Expected Years of Schooling

Example: 13.2

Mean Years of Schooling

Example: 8.5

Gross National Income (GNI) per Capita

Example: 15000

 Predict

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PREDICTION RESULT

High HDI 0.71

This section shows the predicted HDI score or category based on the values entered in the form.

Predict Again

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